

Refractory Poisson: The Effect of the Refractory Period on Neural Rate Coding

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Abstract—The neuron’s membrane potential build-up allows for refractoriness. In this context, the refractory effect is often the primary reason spike trains are not accurately described by traditional counting distributions. Inspired by this biological phenomenon, this paper introduces a counting distribution that includes a parameter specifying the length of the refractory period in neural spike generation. Specifically, the paper considers a lower bound on the intervals between any two spikes that follow an exponential or Gamma distribution. The proposed distribution is a generalized form of standard counting distributions, such as the Poisson distribution, and is named the refractory Poisson distribution. This study aims to provide estimation methods for the proposed distribution. Additionally, it introduces a novel estimation approach called “one-shot” maximum likelihood estimation. The research also provides guidance on the maximum amount of information that can be reliably transferred through a single neuron due to refractoriness. The optimality conditions for the capacity-achieving distribution in the capacity per unit cost problem are presented, which, to the best of our knowledge, have not been reported in the literature. The proposed counting distribution can be used for numbering any random phenomena where a refractory period exists between occurrences. Additionally, the distribution can be applied in various scenarios. For example, it can replace the Poisson distribution in the Generalized Linear Model as an alternative counting distribution.

Index Terms—Computational neuroscience, rate coding, refractory period, count-based distribution, Poisson distribution, parameter estimation, capacity per unit cost.

I. INTRODUCTION

THE purpose of a single neuron is to transmit information about the value of its stimulus intensity to the postsynaptic neurons. To this end, the neuron uses metabolic energy to generate and propagate neural signals. Thus, it is fundamental to understand, at the nodal level of the neuronal network, the relationship between the amount of transferred information and the energy used [1]. The issues raised in the neural coding literature are listed as follows:

- Due to the significant benefits in estimating and validating biological experimental observations, describing the biological neural model with a *single function law*, such as (3), is a key issue in the literature. Therefore,

the descriptors must be analytically resolvable and as representative as possible of neuronal bioactivity [2].

- For the analysis of neural data, an additional step was required after describing the number of spikes occurring within a time window. This estimation step is essential for investigating information processing in the brain and determining the parameters that control the electrical properties of probabilistic biological neurons [3].
- According to the efficient-coding hypothesis [4], another interesting direction is to determine the distribution that achieves the capacity for a noisy neuron. The efficiency of neural communication in terms of reliability has been addressed in [5]. Hence, information theory can be applied to neural systems [6]. The information-theoretic analysis can be extended to account for metabolic expenses. Energy-efficient neural coding was popularized by Levy and Baxter [7]. Specifically, the properties of the neuron are set to minimize energy expenditure for a given amount of information, i.e., the metabolic cost of information [8], or, equivalently, to maximize information transmitted per unit of energy used [9].

By considering the refractory period, the goal of this study is to enhance our understanding of how complex neuronal spike generation mechanisms affect the neuron’s ability to transmit information.

A. Related Work

Here, we divide the literature into two parts. The first part includes models that do not account for refractoriness, while the second part includes models that do.

1) *Neural Models Without Refractory*: [10] and [11] estimated the characteristics of inter-spike intervals (ISIs) by observing individual neuron spike counts, which are independent copies of a doubly Poisson distribution. These characteristics include the mean, coefficient of variation, and the cumulative distribution function (CDF) of ISIs. Regarding temporal coding, [12] considered parameter estimation based on recordings of the potential’s buildup over time, which is modeled by the stochastic differential equation in [13]. Additionally, the study involves the estimation of input intensity. Assuming that ISIs follow a gamma distribution, an explicit form of the capacity-achieving distribution was calculated for an energy-efficient neuron using temporal coding [14]. The study extends to neurons with Generalized Inverse Gaussian (GIG)-distributed ISIs [15], [16]. Ikeda and Manton [17] considered an upper bound on the stimulus intensity in a neuronal

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model with a gamma distribution of ISIs. They proved that, for both rate and temporal coding, the capacity-achieving distribution is discrete with a finite number of support points, and they numerically calculated the distribution. We proved the uniqueness of this capacity-achieving probability distribution using the extended Carleman's condition, which we also formulated in our paper [18]. In [19], we focused on rate coding and introduced the rated inverse Gaussian (rated-IG) distribution, which counts the number of spikes separated by Inverse Gaussian (IG) distributed time periods. The study proved that the capacity-achieving distribution for the rated-IG channel is unique and has finite support. Using the Multi-timescale Adaptive Threshold model of the neuron, energy efficiency was investigated in the context of rate coding [20].

2) *Neural Models With Refractory*: Previous methods of incorporating a refractory period into a stochastic model of spike generation assumed that the *output* firing rate of a neuron is the product of a free firing rate, which depends only on the stimulus, and a recovery function, which describes refractoriness [21]. Based on this assumption, the stochastic model remains the standard Poisson distribution. In studying the role of refractoriness, the authors in [22] and [23] introduced a negative reset potential term, called the *spike response model (SRM)*, to the membrane potential after each spike. They implemented their method using the Peristimulus Time Histogram (PSTH). Richardson incorporated the refractory period into the framework by adding a component to the membrane potential in the form of an integral equation [24]. The effect of the refractory period on neural rate coding has been examined in [25]. However, for simplicity, [25] intuitively considers a uniform distribution as a statistical descriptor, which does not fully capture the effect of the refractory period.

B. Motivations and Challenges

Due to the issues previously listed in the neural coding literature, the motivations and challenges can be categorized as follows:

- In rate coding, traditional counting distributions (e.g., Poisson) ignore the existence of the refractory period, η , which prevents very closely spaced spikes in the neuron. However, due to the refractory period, the ISI cannot be 0 and must be greater than a few milliseconds (i.e., it has a lower bound of η). The inclusion of refractoriness will lead to a more regular [21] and accurate model for describing neuronal behavior. Therefore, a modified counting distribution that incorporates the refractory period is of interest. To the best of our knowledge, the probability distribution of the number of spikes in a time window, considering the assumption of a refractory period, has not yet been discussed in the literature. The situation where a lower bound on the time between consecutive pulses is being considered is both interesting to study and analytically complex. Because the number of events, N , depends on how many previous events have occurred within the time window Δ , the problem becomes non-memoryless. In this context, the probability of a large number of spikes decreases more rapidly compared to the scenario where refractoriness is ignored. This is because

each event that occurs reduces the remaining time window by the length of the refractory period. In other words, the refractory period imposes an upper bound on the number of spikes, N , which cannot be greater than $\frac{\Delta}{\eta} - 1$. We add a spike width of 2ms, in addition to the absolute refractory period of 2ms, resulting in $\eta = 4\text{ms}$ [26]. Hence, by setting $\Delta = 1\text{s}$, the maximum output firing frequency for the biological neuron becomes $\frac{1}{\eta} = 250\text{Hz}$, regardless of the input intensity value. As a consequence, the probability of the output intensity being higher than 250Hz is zero. In such circumstances, a neuron cannot fire too rapidly, and this spiking count belongs to the right-truncated distribution family [27].

- If a realistic counting distribution with refractoriness is available, it becomes desirable to derive the estimation of its parameters. This would provide a reasonably good characterization of the neuron channel.
- It is also worth noting that the problem of optimal information transmission through neurons with refractoriness is still open. In this regard, we are not yet aware of any reports in the literature describing the characteristics of the capacity-achieving distribution. As we are unsure, increasing the refractory period may either increase or decrease the amount of information sent. The same uncertainty applies to the necessary and sufficient conditions in the capacity per unit cost problem.

C. Our Contributions

In the statistics literature, count-based truncated distributions are those that support a subset of positive integers. Truncated distributions are suitable for fitting data when the data originates from a mechanism that excludes the possibility of counting outcomes outside a specific interval. In the literature, the method for truncating a distribution $\Pr(N = n)$ to the interval $[n_l, n_h]$ involves adjusting the distribution by multiplying $\Pr(N = n)$ by a normalizing constant. This multiplication ensures that the sum of all probabilities in the truncated version equals one, as follows:

$$\frac{\Pr(N = n)}{\sum_{n=n_l}^{n=n_h} \Pr(N = n)}; n = n_l, n_l + 1, \dots, n_h, \quad (1)$$

where $\left(\sum_{n=n_l}^{n=n_h} \Pr(N = n)\right)^{-1}$ is the normalizing constant. For all values of $n \in [n_l, n_h]$, the probabilities increase by the same factor, which simplifies the calculations. Interestingly, our proposed distribution is truncated on the right at $\frac{\Delta}{\eta} - 1$ without the need for a normalizing constant. However, this makes the calculations much more challenging. The distribution is properly defined so that its probabilities sum to one (Lemma 1). In line with the motivations previously mentioned, the contributions of this paper are as follows:

- Inspired by the refractory phenomenon, we propose a proper probability mass function (PMF) in a closed-form expression that characterizes the number of events occurring within a time window, assuming the non-refractory portion of the ISIs follows a gamma distribution. The

proposed PMF, obtained by adding a single parameter, dictates the upper bound on the output spike count with respect to the refractory period. The new distribution includes traditional counting distributions as special cases. Next, its properties are discussed. We introduce a new distribution termed *refractory Poisson* for the case of an exponentially distributed non-refractory portion. We hope this proposed counting distribution attracts broader applications in engineering, medicine, economics, and other fields. The distribution can model any random phenomenon with a refractory period between occurrences. Beyond the biological application in neuroscience discussed here, these concepts are applicable to other scenarios. For instance, a refractory period has been observed between successive gene activations [28]. Similarly, the time delay in resetting the magnetic tunnel junction to the P-state mirrors the refractory period [29]. In event-triggered control literature, the focus is often on establishing a positive lower bound on inter-event times to avoid Zeno behavior [30]. Additionally, this distribution could serve as an alternative to the Poisson distribution in the Generalized Linear Model (GLM) as a stochastic spiking process.¹

- We discuss the estimation of the intensity parameter in the proposed PMF using various techniques. The standard maximum likelihood (ML) estimate cannot be expressed in closed form and is therefore obtained numerically. To address this, we propose a new estimation method, resulting in a new closed-form approximation. Extensive simulation studies are conducted to evaluate the performance of these estimates.
- The optimal trade-off between information and energy for a neuron is explored by maximizing the ratio of the amount of information transmitted to the energy used during this transmission, considering the refractory assumption. In this context, we calculate the necessary and sufficient optimality conditions for the optimal distribution in the capacity per unit cost problem and show that optimality is achievable with a discrete distribution for the intensity parameter, consisting of a finite number of mass points. To the best of our knowledge, this is the first time such conditions have been presented. Lastly, the capacity was computed numerically. We demonstrate that increasing the duration of the refractory period enhances the efficiency of the biological neuron in transmitting information, which is consistent with biological experiments [21, Figure 5-F].

D. Organization

The rest of the paper is organized as follows. We provide the essential background of neural signaling in Section II. Section III extends existing counting models by including the refractory period, and the distributional theory and associated properties are derived. In Section IV, three estimation methods are presented for estimating the input parameter from simulated data. In Section V, we justify the discreteness of the capacity-achieving distribution and perform simulations to

numerically calculate the information capacity of the neuron. Finally, the paper concludes in Section VI. The notation used in this paper is as follows: The symbol $\triangleq$ stands for “defined as,” while $\cong$ denotes approximation. The symbol $\sim$ is interpreted as “is distributed as”. $\lceil \cdot \rceil$ denotes the least integer greater than or equal to the argument. Finally, $\Pr(\cdot)$ represents probability, specifically the probability of an event.

II. SPIKE TRAIN PRINCIPLES

This section provides a stochastic framework for understanding how neurons encode input stimuli into spike outputs [32]. It introduces the evolution of the neuron’s membrane potential toward a threshold for spike generation and emphasizes the significance of the refractory period. Stimulus intensity and the underlying neural mechanisms are modeled as random processes that shape spiking activity. We explore neural coding to analyze how refractoriness influences the neuron’s ability to translate input intensity into output spike patterns.

We first introduce the neuron’s resting membrane potential and then explain how synaptic inputs—excitatory or inhibitory—shift this potential toward a threshold. The membrane potential, $V(t)$, is the voltage difference between the neuron’s interior and exterior. In the absence of stimulation, this potential is at a resting level, V_0 . Synaptic inputs can raise or lower $V(t)$, depending on whether they are excitatory or inhibitory. The combined effect of these inputs, known as the postsynaptic potential (PSP), *randomly* moves the membrane potential from its resting state toward a threshold θ . Upon reaching this threshold, the neuron produces an action potential, or spike. The time interval between consecutive spikes is called the inter-spike interval (ISI), denoted as T .

We now describe the refractoriness phenomenon. Immediately after a neuron generates a spike, $V(t)$ returns to V_0 and remains there throughout a period called the absolute refractory period [32]. During this period, which lasts about 2 ms in the primate sensory cortex [14], the sodium channels are inactive and will not open in response to depolarization [33]. The relative refractory period, which coincides with nearly the entire duration of the spike, immediately follows the absolute refractory period and corresponds to hyperpolarization. The term *refractory period*, which lasts for exactly η milliseconds, refers to the combination of the absolute and relative refractory periods.

During the refractory period, it is impossible to change the membrane potential regardless of the intensity of the afferent excitation [34]. Instead, the neuron operates within the non-refractory portion of every inter-spike interval (ISI), denoted as T' . During the non-refractory period, the membrane potential randomly changes over time from the resting potential to the threshold. As implicitly mentioned, the non-refractory period is equal to the first passage time (FPT) of the process, at which $V(t)$ first reaches θ starting from V_0 . To be precise, we divide every ISI duration T into a refractory (i.e., non-active) period and a non-refractory (i.e., active) period such that $T = T' + \eta$ (Fig. 1).

We assume that the amplitude and width of spikes remain constant for each particular neuron type (Fig. 1), consistent

¹One should note, however, that similar to [31], our approach can be used for both GLM and non-GLM likelihood functions.

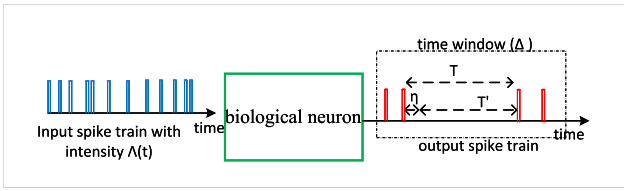


Fig. 1. The intensity $\Lambda(t)$ and the number of spikes in the window are the input and output of the biological neuron, respectively, where η represents the refractory period.

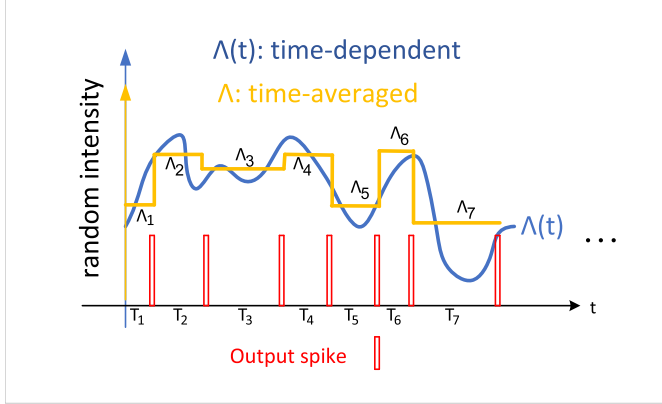


Fig. 2. Continuous intensity $\Lambda(t)$, represented by the blue curve, was averaged across the output ISIs to obtain the discrete Λ , represented by the yellow curve.

with previous literature [14], [15], [17]. The width is typically 2ms [35], which is much smaller than the ISI. Therefore, we regard the spike as an impulse, as stated by previous studies [14], [15], [17]. In this context, the spike is an all-or-nothing pulse, and the spike train of each neuron—that is, the spike sequence—can be represented by a list of the times at which spikes are generated. From a mathematical perspective, the spike train is well-modeled as a point process characterized by a specified intensity.

Each neuron receives multiple spike trains from its neighboring neurons. Assuming statistical independence, these input spike trains are combined to form a single stimulus spike train at the input (the blue spike train in Fig. 1). This resulting spike train is characterized by a time-dependent random intensity, denoted as $\Lambda(t)$. Hence, the neuronal response to the selected stimuli depends on the *intensity of the stimuli*. In line with [14, Theorem 1] and [12], [15], it is common to perform a time-averaging operation over the random process $\Lambda(t)$, as shown by the blue curve in Fig. 2, across the output inter-spike intervals (ISIs) to obtain the random variable Λ , as shown by the yellow curve in Fig. 2. Here, the sample Λ_k represents the time average of $\Lambda(t)$ during the k -th output ISI, as given below:

$$\Lambda_k = \frac{1}{T_k} \int_{T_{k-1}}^{T_k} \Lambda(t) dt.$$

Hence, for each output T_k , the value of Λ_k is defined, and we have the pair (Λ_k, T_k) , where k is the index of the k -th output ISI.

The ISIs are *dependent*, but this dependency is solely through the stimulus intensity. Additionally, the order of this

Markov chain is one. A high or low T_k and T_{k-1} follow each other because their Λ_k and Λ_{k-1} are both high or low, respectively. Mathematically, for the given random sequences $\{T_i\}$ and $\{\Lambda_i\}$, where $i = 1, 2, \dots, I$, it is assumed that the ISI sequences are statistically independent of each other *given* $\Lambda = \lambda$, because the neuron has partially reset its chemical concentrations during the refractory period [14, Eq. 19]:

$$f_{T_1, \dots, T_I | \Lambda_1, \dots, \Lambda_I}(t_1, \dots, t_I | \lambda_1, \dots, \lambda_I) = \prod_{i=1}^I f_{T_i | \Lambda_i}(t_i | \lambda_i).$$

Hence, (Λ_k, T_k) forms a sequence of identically distributed pairs. We suppress the ISI index and simply refer to them as T and Λ , where t_i and λ_i ($i = 1, 2, \dots, I$) are realizations of the random variables T and Λ , respectively. Each value of variable Λ can be estimated by samples of variable T .

The manner in which the intensity of afferent excitation is encoded must be investigated. A higher intensity of afferent excitation increases the likelihood that the neuron will generate a spike sooner. Hence, it is accepted that communication between neurons takes place through spikes [32]. However, there is still some controversy regarding how neurons encode information about the intensity of afferent excitation in different parameters of the efferent spike train [14], [36], [37], [38], [39]. In the rate coding hypothesis, which is the oldest and still one of the most accepted scenarios in neuronal activity analysis [40], information is encoded in the number of spikes, denoted as N , within a given time window, such as Δ . In the temporal coding hypothesis, information is encoded within a single inter-spike interval, T . In both coding hypotheses, a neuron responds to the stimulus intensity, Λ , as its input. It is of interest to determine the effect of refractoriness on rate code efficiency. Therefore, we adopt the first approach in this study.

The neuron was characterized as a noisy information channel, converting the afferent spike train into the efferent spike train. However, as mentioned, random factors such as the opening of an ion channel or the release of a synaptic vesicle mean that one input does not always produce the same response over repeated trials [32]. Hence, the biological neuron corresponds to the notion of a noisy information channel, where the input is afferent excitation intensity, as shown in Fig. 1. How to express the output depends on the coding employed. The output in temporal coding is defined as the time interval between adjacent spikes, while in rate coding, the output is solely the number of spikes observed [17]. Both the dynamics of the stimulus intensity and the random nature of neural mechanisms determine the neuron's output activity. Hence, the neuron's output spike train has a *doubly stochastic property*. In this context, the stimulus intensity Λ is randomly distributed according to a cumulative distribution function (CDF), denoted $F_\Lambda(\lambda)$. Therefore, in rate coding, the neural response is fully described by the conditional probability distribution $\Pr(N = n | \Lambda = \lambda)$, which characterizes the intrinsic nature of the neuron.

In the next section, we introduce a probability distribution that accounts for a minimum separation time between occurrences due to refractoriness.

III. CHANNEL PMF

In this section, we will derive our proposed PMF and explore its related properties. As mentioned, the neuron is viewed as a noisy channel, with the input being the afferent intensity, Λ , and the output being the random efferent spike count, N , within a given time window. Therefore, we are dealing with a counting distribution. A more realistic probability function, $\Pr(N = n|\Lambda = \lambda)$, must be determined.

Assuming each input spike increases $V(t)$ by w , $m = \lceil \frac{\theta}{w} \rceil$ input spikes are required for $V(t)$ to move from the resting potential to the threshold potential θ , resulting in the generation of an output spike. The time required to receive these m spikes corresponds to the sum of m random variables, each following an exponential distribution with intensity λ . Consequently, the time to receive m spikes, allowing the potential to reach the threshold, follows a gamma distribution [14]:

$$f_{T'|\Lambda}(t'|\lambda) = \frac{\lambda^m t'^{m-1} e^{-\lambda t'}}{(m-1)!}, \quad (2)$$

This gamma PDF also results from a first-principle derivation of neuron modeling based on a maximum entropy argument [37]. Additionally, it is one of the most frequently used statistical models in experimental data analysis [41], [42]. Consequently, the time interval between spikes, namely T , is the sum of the gamma-distributed variable T' and the fixed value η . Now, the form of the distribution of N given $\Lambda = \lambda$ is shown in the following theorem, which can be used to investigate the information-theoretical properties of the neuron as an information channel. We name the distribution the *refractory rated-gamma distribution*.

Theorem 1 *refractory rated-gamma distribution: The number of spikes N that occur within a time window Δ , provided that the non-refractory portion of the interval between two subsequent spikes is gamma-distributed in (2), has the following PMF:*

$$\begin{aligned} \Pr(N = n|\Lambda = \lambda) &= e^{-\lambda(\Delta - (n+1)\eta)} \sum_{k=0}^{(n+1)m-1} \frac{\lambda^k (\Delta - (n+1)\eta)^k}{k!} \\ &\quad - e^{-\lambda(\Delta - n\eta)} \sum_{k=0}^{nm-1} \frac{\lambda^k (\Delta - n\eta)^k}{k!}, \end{aligned} \quad (3)$$

which can be expressed compactly using the lower incomplete gamma function [43]:

$$\begin{aligned} \Pr(N = n|\Lambda = \lambda) &= \frac{1}{((n+1)m-1)!} \Gamma((n+1)m, \lambda(\Delta - (n+1)\eta)) \\ &\quad - \frac{1}{(nm-1)!} \Gamma(nm, \lambda(\Delta - n\eta)), \end{aligned} \quad (4)$$

where $n = 0, 1, \dots, \frac{\Delta}{\eta} - 1$, and zero otherwise.

Proof: See Appendix A. ■

We define $h(n) \triangleq \Pr(\sum_{k=0}^n T_k > \Delta)$. Hence, we have $\Pr(N = n|\Lambda = \lambda) = h(n+1) - h(n)$, where

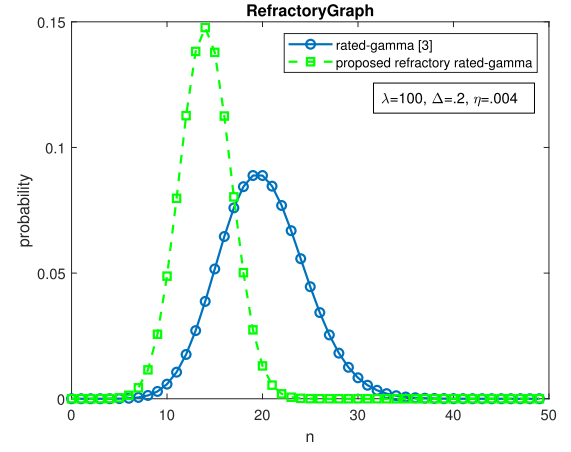


Fig. 3. Graphs of the distribution for specified parameter values are provided. As can be seen, there is a distinct difference when the refractory phenomenon is involved.

$h(n) = e^{-\lambda(\Delta - n\eta)} \sum_{k=0}^{nm-1} \frac{\lambda^k}{k!} (\Delta - n\eta)^k$. If Δ is a multiple of η , the distribution support is $[0, \frac{\Delta}{\eta} - 1]$; otherwise, it is $[0, \lfloor \frac{\Delta}{\eta} \rfloor]$. We consider $\frac{\Delta}{\eta} - 1$ as the maximum value of the support, denoted by n_m , but this is not restrictive. When refractoriness is not considered, i.e., $\eta = 0$, the proposed distribution in (3) reduces to:

$$\Pr(N = n|\Lambda = \lambda) = e^{-\lambda\Delta} \sum_{k=nm}^{(n+1)m-1} \frac{\lambda^k \Delta^k}{k!}, \quad (5)$$

which is mentioned in [17, Lemma 1]. We refer to this reduced distribution as the *rated-gamma distribution*. Under the event-counting hypothesis, the model in (5) is the closest to a realistic channel characterization found in the literature.

In Fig. 3, a simulation study is conducted to evaluate the proposed PMF. In the simulation, the value of stimulus intensity λ is set to 100Hz, based on experimental data from thalamic neurons of macaque monkeys [44], [45]. Fig. 3 highlights the distinction in the probability of a given spike count between the cases of accounting for refractoriness, i.e., the PMF of (3), and not accounting for it, i.e., the PMF of (5). The probability of having a certain number of spikes within a fixed period is inversely related to the number of spikes. The decay rate of the probability increases as the number of spikes rises when refractoriness is considered, compared to when it is ignored in the conventional model. This occurs because, with each event, the time window is reduced by the duration of the refractory period, due to the presence of $n\eta$ in the expression associated with the window, i.e., $(\Delta - n\eta)$, in the proposed PMF rule. In other words, the refractory period imposes an upper bound on the number of events. Regarding the proposed model, we obtain a more accurate statistical parameter comparison to other simpler models (e.g., the model in (5)). We believe this will be of interest to a broad range of scientists.

A. Properties of the Distribution

1) *Proper Distribution:* The following lemma proves that the probabilities add up to one:

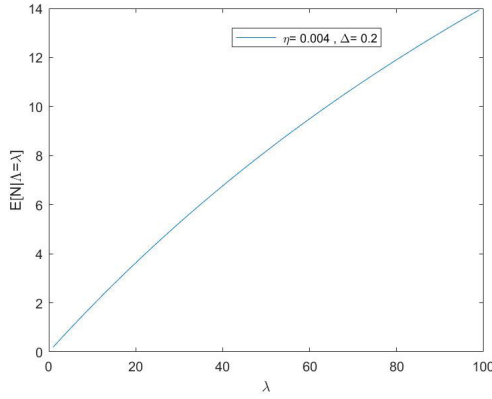


Fig. 4. The mean spike count is plotted against the input intensity.

Lemma 1: Eq. (3) defines a proper PMF, i.e., $\sum_{n=0}^{n_m} \Pr(N = n | \Lambda = \lambda) = 1$.

Proof:

$$\begin{aligned} \sum_{n=0}^{n_m} \Pr(N = n | \Lambda = \lambda) &= \sum_{n=0}^{n_m} (h(n+1) - h(n)) \\ &\stackrel{a}{=} h(n_m + 1) - h(0) \stackrel{b}{=} h(n_m + 1). \end{aligned}$$

The equality *a* follows from the telescoping series approach. The equality *b* holds due to the fact that $h(0) = \Pr\left(\sum_{k=0}^0 T_k > \Delta\right) = 0$. On the other hand, whether n_m is $\frac{\Delta}{\eta} - 1$ or $\left\lfloor \frac{\Delta}{\eta} \right\rfloor$, we have $h(n_m + 1) = \Pr\left(\sum_{k=0}^{n_m+1} T_k > \Delta\right) = 1$. Hence, the proof is complete. ■

2) *Moment:* Here, we discuss the expected value of the refractory rated-gamma distribution in the following theorem:

Theorem 2: The moment $E[N | \Lambda = \lambda]$ can be expressed as:

$$\begin{aligned} E[N | \Lambda = \lambda] &= (n_m + 1) - \sum_{n=0}^{n_m} h(n+1) \\ &= \frac{\Delta}{\eta} - \sum_{n=0}^{n_m} \sum_{k=0}^{(n+1)m-1} e^{-\lambda(\Delta-(n+1)\eta)} \frac{\lambda^k (\Delta - (n+1)\eta)^k}{k!}. \end{aligned} \quad (6)$$

Proof: See Appendix B. ■

The expected value of the response plays a vital role in biological experiments [46]. The plots in Fig. 4 show the variations in $E[N | \Lambda = \lambda]$ for specific parameter values. The pattern of the plots indicates that the curves are monotonically increasing as λ increases and often take on a sigmoidal shape.

3) *Derivative:* In the following, we express the derivative of the function $\Pr(N = n | \Lambda = \lambda)$ with respect to the stimulus intensity λ . This derivative will be used later for the parameter estimation problem.

Lemma 2: The derivative of $\Pr(N = n | \Lambda = \lambda)$ with respect to λ is given below:

$$\begin{aligned} \frac{d\Pr(N = n | \Lambda = \lambda)}{d\lambda} &= e^{-\lambda(\Delta-n\eta)} \frac{\lambda^{nm-1}}{(nm-1)!} (\Delta - n\eta)^{nm} \end{aligned}$$

$$- e^{-\lambda(\Delta-(n+1)\eta)} \frac{\lambda^{(n+1)m-1}}{((n+1)m-1)!} (\Delta - (n+1)\eta)^{(n+1)m}. \quad (8)$$

Proof: See Appendix C. ■

B. Refractory Poisson

By setting $m = 1$ in (3), we obtain:

$$\begin{aligned} \Pr(N = n | \Lambda = \lambda) &= e^{-\lambda(\Delta-(n+1)\eta)} \sum_{k=0}^n \frac{\lambda^k (\Delta - (n+1)\eta)^k}{k!} \\ &\quad - e^{-\lambda(\Delta-n\eta)} \sum_{k=0}^{n-1} \frac{\lambda^k (\Delta - n\eta)^k}{k!}. \end{aligned} \quad (9)$$

This distribution is related to the neuron with an exponentially distributed non-refractory portion. We name this distribution *the refractory Poisson distribution*. The traditional Poisson distribution is a special case of this distribution when the refractory period, η , equals zero. The traditional Poisson distribution is a frequently used archetypal model for neural rate coding [32], [47], [48], making it a suitable choice. However, the Poisson process does not account for the refractory phenomenon. In contrast, our proposed model, the refractory Poisson, includes the refractory phenomenon, making it a more realistic model. According to (9), we can write an approximate version of the refractory Poisson distribution as follows:

$$\Pr(N = n | \Lambda = \lambda) \cong e^{-\lambda(\Delta-(n+1)\eta)} \frac{\lambda^n}{n!} (\Delta - (n+1)\eta)^n. \quad (10)$$

1) *Recurrence Relation:* A recurrence relation for (9) can be derived as follows:

Lemma 3: The numbers $\Pr(N = n | \Lambda = \lambda; \Delta, \eta)$ in (9) satisfy the recurrence relation:

$$\begin{aligned} \Pr(N = n | \Lambda = \lambda; \Delta, \eta) &= \Pr(N = n-1 | \Lambda = \lambda; \Delta - \eta, \eta) \\ &\quad + e^{-\lambda(\Delta-(n+1)\eta)} \frac{\lambda^n}{n!} (\Delta - (n+1)\eta)^n \\ &\quad - e^{-\lambda(\Delta-n\eta)} \frac{\lambda^{n-1}}{(n-1)!} (\Delta - n\eta)^{n-1}. \end{aligned} \quad (11)$$

Proof: See Appendix D. ■

2) *The Mode:* With the help of the above recurrence relation, we derive an approximate expression for the mode of the refractory Poisson distribution in the following lemma. When $\eta = 0$, this expression reduces to the mode of the traditional Poisson distribution, i.e., $\lambda\Delta$.

Lemma 4: An approximate expression for the mode parameter, denoted as n_{mode} , of the refractory Poisson distribution is given by:

$$n_{mode} \cong \left\lfloor \frac{\lambda e^{\lambda\eta} (\Delta - \eta)}{1 + \lambda\eta e^{\lambda\eta}} \right\rfloor. \quad (12)$$

Proof: Using the result of Lemma 3, we write:

$$\begin{aligned} \Pr(N = n | \Lambda = \lambda; \Delta, \eta) - \Pr(N = n-1 | \Lambda = \lambda; \Delta - \eta, \eta) &= e^{-\lambda(\Delta-(n+1)\eta)} \frac{\lambda^n}{n!} (\Delta - (n+1)\eta)^n \end{aligned}$$

$$- e^{-\lambda(\Delta-n\eta)} \frac{\lambda^{n-1}}{(n-1)!} (\Delta-n\eta)^{n-1}.$$

If we consider $\Delta \gg \eta$, we can write: $\Pr(N = n-1|\Lambda = \lambda; \Delta-\eta, \eta) \cong \Pr(N = n-1|\Lambda = \lambda; \Delta, \eta)$. On one hand, in real situations, as mentioned, the conventional values of Δ and μ are 200ms and 2ms, respectively, and the constraint $\Delta \gg \mu$ holds. On the other hand, when considering a time slot identical to the refractory period, there can be at most one spike in each slot. Hence, the approximation is suitable. Therefore, we express:

$$\begin{aligned} & \Pr(N = n|\Lambda = \lambda; \Delta, \eta) - \Pr(N = n-1|\Lambda = \lambda; \Delta, \eta) \\ & \cong e^{-\lambda(\Delta-(n+1)\eta)} \frac{\lambda^n}{n!} (\Delta-(n+1)\eta)^n \\ & - e^{-\lambda(\Delta-n\eta)} \frac{\lambda^{n-1}}{(n-1)!} (\Delta-n\eta)^{n-1}. \end{aligned} \quad (13)$$

Now, we should check the sign of (13) with respect to n . By factoring, we have:

$$e^{-\lambda(\Delta-n\eta)} \frac{\lambda^{n-1}}{(n-1)!} (\Delta-n\eta)^n \left(e^{\lambda\eta} \frac{\lambda(\Delta-(n+1)\eta)^{n+1}}{n(\Delta-n\eta)^n} - 1 \right),$$

or, equivalently,

$$\left(e^{\lambda\eta} \frac{\lambda(\Delta-(n+1)\eta)^{n+1}}{n(\Delta-n\eta)^n} - 1 \right). \quad (14)$$

If we also consider $\Delta \gg \eta$, we should be checking the sign of the function

$$\frac{\lambda e^{\lambda\eta} (\Delta-(n+1)\eta)}{n} - 1. \quad (15)$$

The above function is a homographic function with respect to n , which has a unique root: $n_{mode} = \frac{\lambda e^{\lambda\eta} (\Delta-\eta)}{1+\lambda\eta e^{\lambda\eta}}$. The function is negative when $n \geq n_{mode}$ and positive when $n < n_{mode}$, which concludes the proof. ■

A simulation study is performed to assess the accuracy of the approximate mode of the refractory Poisson distribution (Fig. 5). Based on the simulation results, it is observed that the approximation works well in practice. In the next section, assuming that the neuron follows the proposed distribution from this section, which is the closest model to reality, the intensity parameter is estimated using response samples.

IV. PARAMETER ESTIMATION

As mentioned, the neuron conveys the value of intensity, λ , through its output response. Therefore, estimating the intensity based on observations from the output response will provide insights into the intensity that leads to the output spike train. In this context, this section aims to present estimation methods for λ using M independent and identically distributed random samples $x_1, x_2, \dots, x_M$, based on realizations of the proposed spike count N in (9). Furthermore, let f_i , where $i = 0, 1, \dots, n_m$, represent the observed frequency in the samples corresponding to the i -th element from the alphabet of N , i.e., $\{0, 1, 2, \dots, n_m\}$, such that:

$$M = \sum_{i=0}^{n_m} f_i. \quad (16)$$

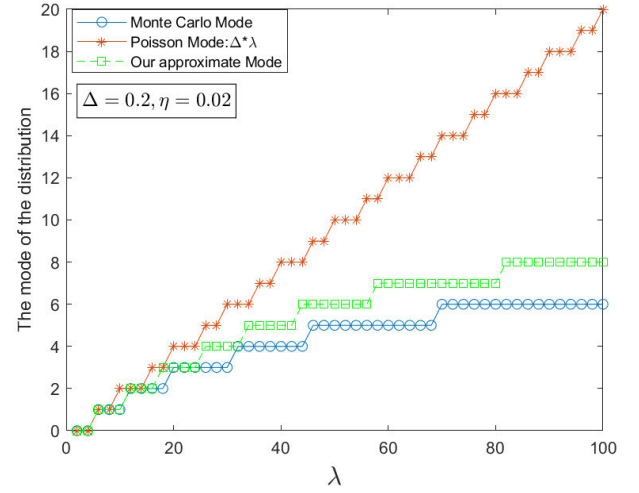


Fig. 5. The approximation mode of the refractory Poisson distribution.

A. Zero-Cell Frequency

Firstly, the method of zero-cell frequency has been considered because studies indicate that this method yields estimators with consistently low efficiencies but simultaneously has less computational complexity and provides a suitable initial estimator [49], [50], [51]. In this regard, it is observed that one or two iterations lead to numerical iteration estimators. Therefore, this method is an interesting alternative to other estimation methods when there are many counts at zero. Let f_0 denote the number of zeros in the observed sample, i.e., $f_0 = \frac{x_0}{M}$. To estimate using the zero-cell frequency method, we set the zero probability in Eq.(9) (i.e., $\Pr(N = 0|\Lambda = \lambda)$) equal to f_0 , as given below:

$$\hat{\lambda}_{ZC} = \frac{1}{\Delta - \eta} \ln \frac{M}{f_0}. \quad (17)$$

In layman's terms, this method calculates the ratio of the number of zero events to the total number of events. This ratio is equal to the probability of zero events in the PMF. The parameters of the probability function are then estimated based on the above equation.

B. Maximum Likelihood

The joint likelihood function corresponding to independently observed variates is given as: $L = \prod_{i=1}^M \Pr(N = x_i|\Lambda = \lambda)$. Hence, the log-likelihood function can be expressed as: $\log L = \sum_{i=1}^M \log \Pr(N = x_i|\Lambda = \lambda)$. By taking the partial derivative of $\log L$ with respect to λ according to Eq. (9), we obtain:

$$\begin{aligned} & \sum_{i=1}^M \frac{\frac{\partial \Pr(N=x_i|\Lambda=\lambda)}{\partial \lambda}}{\Pr(N=x_i|\Lambda=\lambda)} \\ & = \sum_{i=1}^M \frac{\frac{\lambda^{x_i-1}}{(x_i-1)!} (\Delta-x_i\eta)^{x_i} - e^{\lambda\eta} \frac{\lambda^{x_i}}{x_i!} (\Delta-(x_i+1)\eta)^{x_i+1}}{e^{\lambda\eta} \sum_{k=0}^{x_i} \frac{\lambda^k}{k!} (\Delta-(x_i+1)\eta)^k - \sum_{k=0}^{x_i-1} \frac{\lambda^k}{k!} (\Delta-x_i\eta)^k}, \end{aligned} \quad (18)$$

or, equivalently,

$$\sum_{i=0}^{n_m} f_i \left(\frac{\frac{\lambda^{i-1}}{(i-1)!} (\Delta - i\eta)^i - e^{\lambda\eta} \frac{\lambda^i}{i!} (\Delta - (i+1)\eta)^{i+1}}{e^{\lambda\eta} \sum_{k=0}^i \frac{\lambda^k}{k!} (\Delta - (i+1)\eta)^k - \sum_{k=0}^{i-1} \frac{\lambda^k}{k!} (\Delta - i\eta)^k} \right). \quad (19)$$

The solution to the equation obtained by setting the above function, denoted by $q(\lambda)$, equal to zero is the well-known maximum likelihood (ML) estimate of the parameter, denoted by $\hat{\lambda}_{ML}$. For a given dataset, the equation does not provide an explicit ML estimator, so an iterative scheme, such as the Newton-Raphson method, can be applied, as no major simplifications are possible. The Newton-Raphson method is an iterative approach for solving equations. It updates the solution using the derivatives of the function. So, the derivative of the function $q(\lambda)$ with respect to the variable λ , denoted by $q'(\lambda)$, is given in equation (20), as shown at the bottom of the next page. Hence, computing these derivatives adds complexity to the method. While the ML method provides accurate estimations, its disadvantage lies in requiring highly complex computations and lacking a closed-form solution. The uniqueness of the solution to the equation has not been demonstrated in this paper. However, in all the simulation work conducted, the equation consistently yields a unique solution. Since the ML method may not converge to a solution, we use the initial estimate obtained from the zero-cell frequency method. Given the maximization equation and its derivative in (19) and (20), respectively, a pseudocode algorithm is provided in Algorithm 1 for implementing the Newton-Raphson method.

Algorithm 1 Pseudocode for Newton-Raphson Method

```

1: Input:  $q(\lambda)$ ,  $q'(\lambda)$ ,  $\hat{\lambda}_{ZC}$ ,  $\text{tol} = 10^{-4}$ ,  $\text{max\_iter} = 100$ 
2:  $\hat{\lambda}_{ML} \leftarrow \hat{\lambda}_{ZC}$ 
3: for iteration = 1 to  $\text{max\_iter}$  do
4:   Compute  $q(\hat{\lambda}_{ML})$  and  $q'(\hat{\lambda}_{ML})$ 
5:    $\hat{\lambda}_{ML-new} \leftarrow \hat{\lambda}_{ML} - \frac{q(\hat{\lambda}_{ML})}{q'(\hat{\lambda}_{ML})}$ 
6:   if  $|\hat{\lambda}_{ML-new} - \hat{\lambda}_{ML}| < \text{tol}$  then
7:     Print: " $\lambda_{ML-new}$ "
8:     Exit
9:   end if
10:   $\hat{\lambda}_{ML} \leftarrow \hat{\lambda}_{ML-new}$ 
11: end for

```

C. Proposed Estimation Method

The ML approach is based on finding the parameter that maximizes the joint probability function of observed variates. In most cases, the likelihood equation is too complex (e.g., Eq. (19)) to yield a closed-form solution, so the estimate must be evaluated numerically. However, it is sometimes possible to obtain a closed-form estimate of the parameter that maximizes a single probability function. In this context, when a rapid estimation method without iteration is desired, or when reasonable initial estimates for iterative procedures are

needed, we propose a method motivated by intuitive reasoning. We have named this method one-shot maximum likelihood (OS-ML). Since the closed-form derivative does not exist in its general form with M samples, we propose using a representative sample as an approximate solution. OS-ML method addresses this by using the average of the available samples as a representative of the individual sample. In fact, the sample mean is the most “expected” value of a random variable because the expected-value of a random variable from its mean is zero. In this method, we maximize the probability

of the sample mean, $\bar{x} = \frac{\sum_{i=1}^M x_i}{M}$, instead of the joint probability, as follows:

$$\hat{\lambda}_{OS-ML} = \arg \max_{\lambda} \Pr \left(\frac{\sum_{i=1}^M x_i}{M}; \Lambda = \lambda \right). \quad (21)$$

If we apply the method to the traditional Poisson distribution, we obtain the following:

$$\frac{\partial}{\partial \lambda} \left(\frac{e^{-\lambda} \lambda^{\bar{x}}}{\bar{x}!} \right) = 0 \Rightarrow \hat{\lambda}_{OS-ML} = \bar{x}. \quad (22)$$

which coincides with the standard ML estimate. The same outcome occurs for the rate parameter of the Gamma distribution, further corroborating our choice of a representative sample. If we apply the method to the refractory Poisson distribution, we obtain the following:

$$\frac{\partial}{\partial \lambda} (\Pr(N = \bar{x} | \Lambda = \lambda)) = 0. \quad (23)$$

Plugging (8) into the above equation and after some algebra, we get:

$$e^{\eta \hat{\lambda}_{OS-ML}} \hat{\lambda}_{OS-ML} = \bar{x} \frac{(\Delta - \bar{x}\eta)^{\bar{x}}}{(\Delta - (\bar{x} + 1)\eta)^{\bar{x}+1}}. \quad (24)$$

or, approximately,

$$\hat{\lambda}_{OS-ML} = \bar{x} \frac{(\Delta - \bar{x}\eta)^{\bar{x}}}{(\Delta - (\bar{x} + 1)\eta)^{\bar{x}+1}}. \quad (25)$$

Therefore, the OS-ML method offers the advantage of providing a closed-form estimation; however, it results in an approximate estimate.

D. Simulation Experiments

Fig. 6 illustrates an extensive simulation study conducted to evaluate the performance of different estimators. Additionally, we provide some numerical results in the inset image to demonstrate the performance of the OS-ML method, which indicates the distance between the true lambda and the estimated lambda for both ML and OS-ML. The study is based on 1,000 simulated data sets. The results of the simulation experiments indicate that the estimators perform quite well in practice.

In the next section, assuming that the neuron follows the proposed distribution from this paper, which accounts for the refractory phenomenon, we explore how the neuron should operate to optimally transmit information.

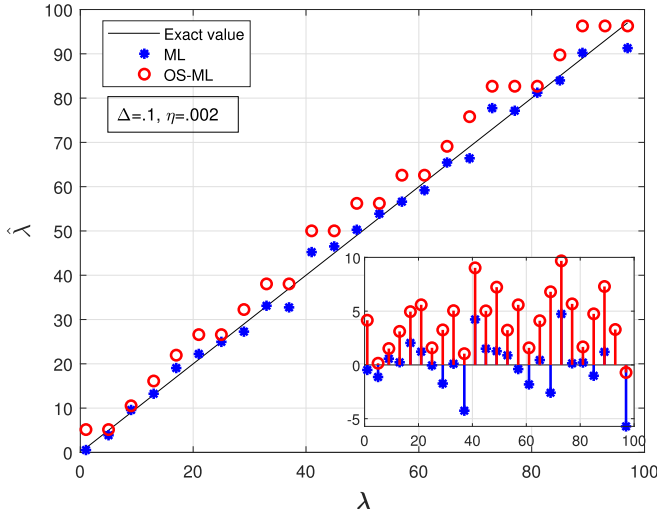


Fig. 6. The behavior of different estimation methods. The inset image demonstrates the distance between the true lambda and the estimated lambda for both ML and OS-ML.

V. MAXIMIZING INFORMATION PER UNIT COST

This section addresses the challenge of determining the stimulus intensity distribution that maximizes the capacity of a noisy biological channel [52]. For a channel governed by the proposed probability distribution in (3), which includes refractory effects, we formulate the problem as a capacity-per-unit-cost optimization. Without relying on traditional transformations, we derive conditions for the optimal input distribution. Our results show that refractory effects enhance neural capacity, with the discrete optimal input distribution aligning with observed binary encoding in neuronal systems.

The information capacity at the cellular level paves the way toward computing the brain's overall information capacity, which provides an upper bound on the information about stimulus intensity that can be reliably transferred [14]. However, according to the efficient coding hypothesis [7] and other theories, it is believed that the rate of information transfer in biological neurons is close to this capacity. Therefore, the concept of capacity is useful in the context of neuronal systems. In this framework, a neuron maximizes Shannon mutual information, $I(\Lambda; N)$, transmitted per ATP molecule, by allocating optimal probabilities to the intensity of the input spike train. Mutual information quantifies the amount of information one part of a system contains that can explain another part.

For biological plausibility, Λ is restricted to an interval, say $[a, b]$. Another natural assumption is that Λ is randomly chosen

from a CDF, denoted by $F_\Lambda(\lambda)$. The marginal information density is defined as [53]: $i(\lambda; F_\Lambda(\lambda)) \triangleq \sum_{n=0}^{n_m} \Pr(N = n | \Lambda = \lambda) \log \frac{\Pr(N=n|\Lambda=\lambda)}{\int_a^b \Pr(N=n|\Lambda=\lambda) dF_\Lambda(\lambda)}$. As a consequence, the mutual information is formulated as follows, expressed as the expectation of the marginal information density relative to $F_\Lambda(\lambda)$:

$$I(\Lambda; N) = \int_a^b i(\lambda; F_\Lambda(\lambda)) dF_\Lambda(\lambda) = \mathbb{E}_\Lambda [i(\lambda; F_\Lambda(\lambda))] \triangleq I(F_\Lambda(\lambda)).$$

Nervous systems maximize performance while minimizing metabolic energy expenditure [54]. Adopting the rate coding hypothesis, [55] showed that signaling-related energy consumption increases linearly with the mean spiking count, i.e., $\mathbb{E}_N[\kappa N]$, where $\kappa (= 7 \times 10^8 \text{ ATP molecules})$ represents the metabolic cost of generating a single spike. Since N depends on the intensity Λ , the metabolic cost of neuronal activity is linked to Λ as follows [45]:

$$g(\lambda) \triangleq \mathbb{E}_N[\kappa N | \Lambda = \lambda]. \quad (26)$$

The average metabolic cost associated with neuronal variability is $\mathbb{E}_\Lambda[\mathbb{E}_N[\kappa N | \Lambda = \lambda]] \triangleq G(F_\Lambda(\lambda))$. When the energy-efficiency criterion is imposed, the optimal trade-off between transmitted information and expended energy is defined by the capacity per unit cost problem [7, Eq. (3.6)]:

$$\max_{F_\Lambda(\lambda)} \frac{\int_a^b \left(\sum_{n=0}^{n_m} \Pr(N = n | \Lambda = \lambda) \log \frac{\Pr(N=n|\Lambda=\lambda)}{\int_a^b \Pr(N=n|\Lambda=\lambda) dF_\Lambda(\lambda)} \right) dF_\Lambda(\lambda)}{\mathbb{E}_\Lambda[\mathbb{E}_N[\kappa N | \Lambda = \lambda]]}. \quad (27)$$

whose solution, denoted as $F_0(\lambda)$, is the capacity-achieving input distribution. The important point is that the solution in (27) to the optimization problem is a *function*, not a single value. In this case, the following basic optimization theorem provides the necessary and sufficient conditions (known as the KKT conditions) for the solution [53]:

Theorem 3: *From a compact and convex topological space $\Omega(F(x))$, let $J(F(x))$ be a weak continuous, weakly-differentiable, and concave and real-valued functional related to $F(x)$. Then the maximum $C = \sup_{F(x) \in \Omega(F(x))} J(F(x))$ exists for some $F_0(x)$. If J is strictly concave, the maximum is achieved by a **unique** solution. A necessary and sufficient condition for $J(F_0(x)) = C$ is that $J_{F_0(x)}'(F(x)) \leq 0$ for all*

$$q'(\lambda) \triangleq \sum_{i=0}^{n_m} f_i \left(\frac{\frac{\lambda^{i-2}}{(i-2)!} (\Delta - i\eta)^i - \frac{\eta e^{\eta\lambda} \lambda^i}{i!} (\Delta - (i+1)\eta)^{i+1} - \frac{e^{\eta\lambda} \lambda^{i-1}}{i-1!} (\Delta - (i+1)\eta)^{i+1}}{e^{\eta\lambda} \sum_{k=0}^i \frac{\lambda^k}{k!} (\Delta - (i+1)\eta)^k e^{\eta\lambda} \sum_{k=0}^{i-1} \frac{\lambda^k}{k!} (\Delta - i\eta)^k} - \left[\frac{\frac{\lambda^{i-1}}{i-1!} (\Delta - i\eta)^i e^{\eta\lambda} \frac{\lambda^i}{i!} (\Delta - i\eta)^{i+1}}{e^{\eta\lambda} \sum_{k=0}^i \frac{\lambda^k}{k!} (\Delta - (i+1)\eta)^k - \sum_{k=0}^{i-1} \frac{\lambda^k}{k!} (\Delta - i\eta)^k} \right]^2 \right). \quad (20)$$

F in $\Omega(F(x))$, where $J_{F_0(x)}'(F(x))$ is the weak derivative of J in F at F_0 , i.e.,

$$J_{F_0}(F(x)) \triangleq \lim_{\varepsilon \rightarrow 0} \frac{J((1-\varepsilon)F_0(x) + \varepsilon F(x)) - J(F_0(x))}{\varepsilon}. \quad (28)$$

Proof: See [56, Sec. 7.4, Th. 2]. ■

The capacity-per-unit-cost problem mentioned in (27) is rarely solvable analytically due to the difficulty in determining the optimality conditions of the objective function, which is expressed as a fraction with a numerator and a denominator [57]. Instead, the literature typically transforms (27) into the capacity-cost function based on [57, Theorem 2]. In the capacity-cost function, a Lagrange multiplier parameter is added to the objective function and subsequently to the optimality conditions. However, this multiplier is unknown beforehand, which can present a significant challenge. Additionally, another optimization problem must be solved where the cost is the optimization variable [57, Theorem 2]. Previous studies [16], [45], [46], [53], [58], [59], [60], [61], [62] did not perform the calculations related to the weak derivation of the objective function for the capacity per unit cost problem and instead converted it to the capacity-cost function according to Theorem 2 in [57].

In this paper, for the first time, we do not convert the capacity per unit cost problem into the capacity-cost function. Instead, we directly derive the optimality conditions from the objective function of the capacity per unit cost problem. We derive the KKT conditions for this scenario in the following theorem, which no longer includes the Lagrange multiplier, making it easier to verify whether the conditions are satisfied.

Theorem 4 KKT conditions: Let E_0 denote the points of increase of CDF $F_0(\lambda)$ on $\lambda \in [a, b]$, i.e., the points with a non-zero probability. Capacity per unit cost in (27) is achieved by $F_0(\lambda)$ if and only if:

$$\lambda \in E_0 : I(F_0(\lambda))g(\lambda) - G(F_0(\lambda))i(N; F_0(\lambda)) = 0. \quad (29)$$

$$\lambda \notin E_0 : I(F_0(\lambda))g(\lambda) - G(F_0(\lambda))i(N; F_0(\lambda)) > 0. \quad (30)$$

Proof: We rewrite the objective function (27) as follows:

$$J = \frac{I(F_\Lambda(\lambda))}{G(F_\Lambda(\lambda))}. \quad (31)$$

Next, we establish the weak differentiability of the above ratio with respect to $F_\Lambda(\lambda)$ at $F_0(\lambda)$ in Eq. (32), as shown at the bottom of the next page. Now, according to Theorem 3, we set $J'_{F_0}(F) < 0$, i.e.,

$$I'_{F_0}(F)G(F_0(\lambda)) - G'_{F_0}(F)I(F_0(\lambda)) < 0. \quad (33)$$

Moreover, we have:

$$I'_{F_0}(F) = \int_a^b i(N; F_\Lambda(\lambda)) dF_\Lambda(\lambda) - \int_a^b i(N; F_\Lambda(\lambda)) dF_0(\lambda), \quad (34)$$

and

$$G'_{F_0}(F) = \int_a^b g(\lambda) dF_\Lambda(\lambda) - \int_a^b g(\lambda) dF_0(\lambda). \quad (35)$$

By substituting (34) and (35) into (33), we obtain the necessary and sufficient conditions for the solution $F_0(\lambda)$ as follows:

$$\int_a^b [i(N; F_\Lambda(\lambda))G(F_0(\lambda)) - g(\lambda)I(F_0(\lambda))] dF_\Lambda(\lambda) \leq 0; \forall F_\Lambda(\lambda).$$

Checking the above condition for optimality is impossible because it requires verifying the validity of the relation for $\forall F_\Lambda(\lambda)$ to determine the optimality of $F_0(\lambda)$. However, by following the approach of [16], [53], [58], [59], [60], [61], and [62], we can prove that the above condition holds if and only if the conditions in Theorem 4 are satisfied. Therefore, the proof is complete. ■

On the one hand, the complexity of the proposed PMF prevents us from calculating $F_0(\lambda)$ in closed form due to the condition in Theorem 4. On the other hand, obtaining a numerical $F_0(\lambda)$ is impossible because of the infinite search space. In what follows, we discuss a property of $F_0(\lambda)$ that aids in calculating the optimal distribution numerically. Since the number of spikes as channel output is limited, i.e., the output alphabet in (3) is finite, Dubins' Theorem [63] guarantees that the optimal input distribution is discrete and contains no more than $n_m + 1$ mass points. Thus, the capacity-achieving distribution belongs to a finite-dimensional space. In biological sensory systems, the input intensity Λ is a stochastic function of the external sensory stimulus value x , expressed as $\Lambda = g(x)$, where $g(\cdot)$ is known as the tuning curve [48]. This is significant for biological experimental studies, as it implies that information is maximally transmitted when the stimuli to the neuron have discrete states. However, the number of mass points l , the positions $a \leq \lambda_1 < \lambda_2 < \dots < \lambda_l \leq b$, and the probabilities $0 < p_{\lambda_1} \leq p_{\lambda_2} \leq \dots \leq p_{\lambda_l} < 1$ (with $\sum_{i=1}^l p_{\lambda_i} = 1$) are still unknown and are part of the optimization parameters. Therefore, the capacity per unit cost in (27) is rewritten as:

$$\max_{F_\Lambda(\lambda)} \frac{\sum_{i=1}^l p_{\lambda_i} \sum_{j=0}^{n_m} \Pr(N = n_j | \Lambda = \lambda_i) \log \frac{\Pr(N = n_j | \Lambda = \lambda_i)}{\sum_{k=1}^l p_{\lambda_k} \Pr(N = n_j | \Lambda = \lambda_k)}}{\sum_{i=1}^l p_{\lambda_i} E_N[\kappa N | \Lambda = \lambda]}.$$

A numerical method, such as the Jimbo-Kunisawa algorithm [64], can now be employed due to the conditions of Theorem 4. This method is an extension of the Blahut-Arimoto algorithm, designed to calculate the channel capacity per unit cost. It iteratively optimizes the *ratio* of capacity to cost, making it applicable in situations where input costs vary. If a distribution satisfies the KKT condition, then the global optimal solution is found. We calculate the optimal distribution and its KKT condition for an empirical model of a neuron ($\Delta = 0.1$ s, $\eta = 0.002$ s, $a = 5$ Hz and $b = 20$ Hz). For details on the numerical simulations, see Algorithm 2, where the optimization parameters are alternately updated with respect to the set of parameters until the KKT conditions for optimality are satisfied.

Fig. 7 shows that the KKT conditions hold, confirming that the optimal input distribution exists uniquely and is discrete,

Algorithm 2 Outline of Programming Procedure**Input:** $l=2$;

- 1: Initialize the position (i.e., λ_i) and probability (i.e., $p(\lambda_i)$), where $1 \leq i \leq l$.
- 2: Fix the position and optimizing the probability with Jimbo-Kunisawa algorithm
- 3: Fix the probability and optimizing the positions by using a gradient-based algorithm.
- 4: If convergence of $p(\lambda_i)$ and λ_i not met, go to Step 2.
- 5: Discretize λ and evaluate plot $i(\lambda)$ for the probability mass and other masses. If KKT condition (from Theorem 4) is violated, then $l = l + 1$ and goto Step 1.
- 6: **return** position, probability

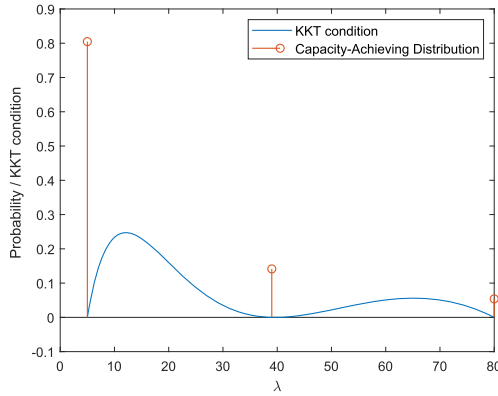


Fig. 7. The calculated distribution satisfies the KKT conditions, indicating the optimality of the distribution.

supported at only three points with nonzero probability. Fig. 8 highlights how the probabilities and locations of the mass points of the optimal distribution evolve with increasing input intensity b , as shown in [65, Fig. 2]. In this figure, we varied the value of b from 20 Hz to 100 Hz. Two technical points can be inferred from the figure. The first point is that the capacity-achieving distribution always has a mass point at the beginning of the interval. This has been proven for simple distributions (e.g., Exponential in [58] and Poisson in [65]). The second point is that the simulations indicate that using neurons as a binary code is a more efficient approach, where only two states are possible. This optimality of binary encoding

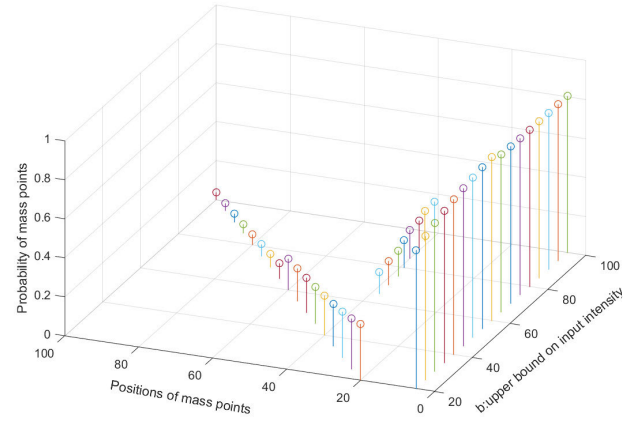


Fig. 8. Evolving the probabilities and locations of the mass points of the optimal distribution with increasing upper bounds of input intensity.

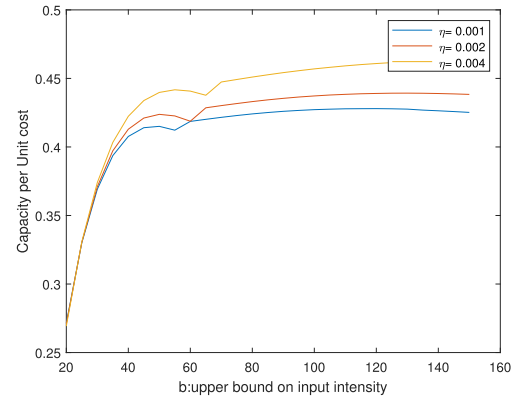


Fig. 9. Capacity per unit cost is plotted against the upper bound on input intensity. This figure illustrates the fundamental point that as the refractory period increases, the amount of information per unit cost transmitted by the biological neuron increases.

is consistent with experimental studies [66]. As the upper bound increases, the capacity-achieving distribution approximately becomes ternary. In Fig. 9, capacity per unit cost is plotted against the upper bound on input intensity. We also observe that the refractory period phenomenon increases the number of bits transmitted, which is consistent with [21, Figure 5-F].

$$\begin{aligned}
 J'_{F_0}(F) &= \lim_{\varepsilon \rightarrow 0} \frac{\frac{I((1-\varepsilon)F_0(\lambda) + \varepsilon F_\Lambda(\lambda))}{G((1-\varepsilon)F_0(\lambda) + \varepsilon F_\Lambda(\lambda))} - \frac{I(F_0(\lambda))}{G(F_0(\lambda))}}{\varepsilon} \\
 &= \lim_{\varepsilon \rightarrow 0} \frac{I((1-\varepsilon)F_0(\lambda) + \varepsilon F_\Lambda(\lambda))G(F_0(\lambda)) - G((1-\varepsilon)F_0(\lambda) + \varepsilon F_\Lambda(\lambda))I(F_0(\lambda))}{\varepsilon G((1-\varepsilon)F_0(\lambda) + \varepsilon F_\Lambda(\lambda))G(F_0(\lambda))} \\
 &= \lim_{\varepsilon \rightarrow 0} \frac{\frac{I((1-\varepsilon)F_0(\lambda) + \varepsilon F_\Lambda(\lambda))G(F_0(\lambda))}{\varepsilon} - \frac{G((1-\varepsilon)F_0(\lambda) + \varepsilon F_\Lambda(\lambda))I(F_0(\lambda))}{\varepsilon}}{G((1-\varepsilon)F_0(\lambda) + \varepsilon F_\Lambda(\lambda))G(F_0(\lambda))} \\
 &= \frac{I'_{F_0}(F)G(F_0(\lambda)) - G'_{F_0}(F)I(F_0(\lambda))}{G(F_0(\lambda))^2}.
 \end{aligned} \tag{32}$$

VI. CONCLUSION AND DISCUSSIONS

We proposed a probability distribution capable of describing count data with strong physiological motivation. This distribution accounts for a refractory period, i.e., a lower bound between occurrences, and encompasses special cases of the traditional Poisson distribution. Next, we derived relevant statistical properties. We hope that this generalization may attract wider applications in engineering, medicine, economics, and other fields. We also discussed parameter estimation using different methods and conducted several simulation studies to evaluate the performance of the proposed methods. Concerning the refractory property, we found that maximizing the input distribution consists of a finite number of mass points and explored how optimal information transmission deals with metabolic cost. Additionally, we derived the necessary and sufficient conditions for the capacity per unit cost problem. Further work could involve proving the log-concave property for the proposed PMF. Additionally, future research could focus on obtaining an approximation and lower bounds on capacity, or investigating a group of coupled neurons.

APPENDIX A

PROOF OF THE PMF EQUATION

Because $N = n$ if and only if $\sum_{k=1}^n T_k \leq \Delta < \sum_{k=1}^{n+1} T_k$, we write:

$$\begin{aligned} \Pr(N = n | \Lambda = \lambda) &= \Pr\left(\sum_{k=1}^n T_k \leq \Delta < \sum_{k=1}^{n+1} T_k\right) \\ &= \Pr\left(\sum_{k=1}^n T_k \leq \Delta\right) - \Pr\left(\sum_{k=1}^n T_k \leq \Delta \cap \sum_{k=1}^{n+1} T_k \leq \Delta\right) \end{aligned}$$

$$\begin{aligned} &= \Pr\left(\sum_{k=1}^n T_k \leq \Delta\right) - \Pr\left(\sum_{k=1}^{n+1} T_k \leq \Delta\right) \\ &\stackrel{a}{=} \Pr\left(\sum_{k=1}^{n+1} T_k \geq \Delta\right) - \Pr\left(\sum_{k=1}^n T_k \geq \Delta\right) \\ &\stackrel{b}{=} \Pr\left(\sum_{k=1}^{n+1} T'_k \geq \Delta - (n+1)\eta\right) - \Pr\left(\sum_{k=1}^n T'_k \geq \Delta - n\eta\right). \end{aligned} \quad (36)$$

The equality a is due to the complement rule from the probability axioms. The equality b is derived from $T = T' + \eta$. Using the fact that the sum of independent gamma-distributed variables is itself gamma-distributed, $\sum_{k=1}^{n+1} T'_k \sim \text{Gamma}((n+1)m, \lambda)$ and $\sum_{k=1}^n T'_k \sim \text{Gamma}(nm, \lambda)$. Applying the survival function of the gamma distribution to Eq. (36) completes the proof.

APPENDIX B

PROOF OF THE MOMENT EQUATION

we have:

$$\mathbb{E}[N | \Lambda = \lambda] = \sum_{n=0}^{n_m} n \Pr(N = n | \Lambda = \lambda) \quad (37)$$

$$= \sum_{n=0}^{n_{\max}} n [h(n+1) - h(n)] = \sum_{n=0}^{n_m} n h(n+1) - \sum_{n=0}^{n_m} n h(n) \quad (38)$$

$$= \sum_{n=0}^{n_m} (n+1) h(n+1) - \sum_{n=0}^{n_m} n h(n) - \sum_{n=0}^{n_m} h(n+1) \quad (39)$$

$$\begin{aligned} &\frac{d \Pr(N = n | \Lambda = \lambda)}{d\lambda} \\ &= -(\Delta - (n+1)\eta) e^{-\lambda(\Delta - (n+1)\eta)} \sum_{k=0}^{(n+1)m-1} \frac{\lambda^k}{k!} (\Delta - (n+1)\eta)^k \end{aligned} \quad (41)$$

$$\begin{aligned} &+ e^{-\lambda(\Delta - (n+1)\eta)} \sum_{k=1}^{(n+1)m-1} \frac{k \lambda^{k-1}}{k!} (\Delta - (n+1)\eta)^k + (\Delta - n\eta) e^{-\lambda(\Delta - n\eta)} \sum_{k=0}^{nm-1} \frac{\lambda^k}{k!} (\Delta - n\eta)^k \\ &- e^{-\lambda(\Delta - n\eta)} \sum_{k=1}^{nm-1} \frac{k \lambda^{k-1}}{k!} (\Delta - n\eta)^k \end{aligned}$$

$$\begin{aligned} &= -(\Delta - (n+1)\eta) e^{-\lambda(\Delta - (n+1)\eta)} \sum_{k=0}^{(n+1)m-1} \frac{\lambda^k}{k!} (\Delta - (n+1)\eta)^k \\ &+ (\Delta - (n+1)\eta) e^{-\lambda(\Delta - (n+1)\eta)} \sum_{k=0}^{(n+1)m-2} \frac{\lambda^k}{k!} (\Delta - (n+1)\eta)^k \end{aligned} \quad (42)$$

$$+ (\Delta - n\eta) e^{-\lambda(\Delta - n\eta)} \sum_{k=0}^{nm-1} \frac{\lambda^k}{k!} (\Delta - n\eta)^k - (\Delta - n\eta) e^{-\lambda(\Delta - n\eta)} \sum_{k=0}^{nm-2} \frac{\lambda^k}{k!} (\Delta - n\eta)^k \quad (43)$$

$$= -e^{-\lambda(\Delta - (n+1)\eta)} \frac{\lambda^{(n+1)m-1}}{((n+1)m-1)!} (\Delta - (n+1)\eta)^{(n+1)m} + e^{-\lambda(\Delta - n\eta)} \frac{\lambda^{nm-1}}{(nm-1)!} (\Delta - n\eta)^{nm}. \quad (44)$$

$$\begin{aligned}
& \stackrel{a}{=} (n_m + 1)h(n_m + 1) - \sum_{n=0}^{n_m} h(n + 1) \\
& = \frac{\Delta}{\eta} - \sum_{n=0}^{n_m} h(n + 1), \tag{40}
\end{aligned}$$

where equality a is due to the telescoping series approach. Eq. (40) cannot be further simplified. This concludes the proof.

APPENDIX C

THE DERIVATIVE OF THE PMF EQUATION

From the PMF in (3), we write as given in (44), as shown at the bottom of the previous page.

APPENDIX D

PROOF OF THE RECURRENCE EQUATION

From the PMF in (9), we have:

$$\begin{aligned}
& \Pr(N = n | \Lambda = \lambda; \Delta, \eta) \\
& = e^{-\lambda(\Delta - \eta - n\eta)} \sum_{k=0}^{n-1} \frac{\lambda^k}{k!} (\Delta - \eta - n\eta)^k \\
& \quad + e^{-\lambda(\Delta - (n+1)\eta)} \frac{\lambda^n}{n!} (\Delta - (n+1)\eta)^k \\
& \quad - e^{-\lambda(\Delta - \eta - (n-1)\eta)} \sum_{k=0}^{n-2} \frac{\lambda^k}{k!} (\Delta - \eta - (n-1)\eta)^k \tag{45} \\
& \quad - e^{-\lambda(\Delta - n\eta)} \frac{\lambda^{n-1}}{(n-1)!} (\Delta - n\eta)^{n-1} \\
& = \Pr(N = n - 1 | \Lambda = \lambda; \Delta - \eta, \eta) \\
& \quad + e^{-\lambda(\Delta - (n+1)\eta)} \frac{\lambda^n}{n!} (\Delta - (n+1)\eta)^n \tag{46} \\
& \quad - e^{-\lambda(\Delta - n\eta)} \frac{\lambda^{n-1}}{(n-1)!} (\Delta - n\eta)^{n-1}.
\end{aligned}$$

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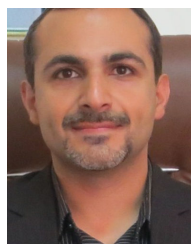
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